Household Wealth and Entrepreneurial Career Choices: Evidence from Climate Disasters

by

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Abstract

This study investigates how household wealth affects the human capital of startups, based on U.S. Census individual-level employment data, deed records, and geographic information system (GIS) data. Using floods as a wealth shock, a regression discontinuity analysis shows inundated residents are 7% less likely to work in startups relative to their neighbors outside the flood boundary, within a 0.1-mile-wide band. The effect is more pronounced for homeowners, consistent with the wealth effect. The career distortion leads to a significant long-run income loss, highlighting the importance of self-insurance for human capital allocation.

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1. Introduction

Human capital is a central input to the success of firms (Schultz, 1961). Startup firms, an important driver of economic growth and innovation, tend to rely heavily on the human capital provided by early-stage employees. According to the State of Startups survey (First Round Capital, 2017), the top concern of new business founders is attracting qualified employees. Choi et al. (2021) estimates that losing an early-stage worker can hamper the long-run revenue growth by 15%–30% for a startup firm. Despite the importance of human capital to startups, we have a limited understanding of individuals' decisions to pursue startup jobs. Previous studies mainly focus on the decision to found new businesses (e.g., Hurst and Lusardi, 2004). However, these findings may not necessarily apply to understanding the decision to work for startups (e.g., Roach and Sauermann, 2015).

According to the theoretical literature, a key input in individual employment decisions is household wealth (e.g., Danforth, 1979). This paper aims to investigate how household wealth affects individuals' decision to work in startups, using climate disasters as an exogenous shock to household wealth. My empirical analysis builds on a novel integration of flood data in geographic information system (GIS) format, U.S. Census individual-level employment data, and deed records. The key inputs are a set of staggered floods in the U.S. that I use as household wealth shocks. For each flood, there were households just above and below the actual inundation line, although they were ex-ante indistinguishable. The subsequent career choices made by the two groups within a narrow band (0.1 miles wide) in the same community can be reliably attributed to the impact of the flood damage, which can be viewed as an exogenous shock to household wealth.

A regression discontinuity (RD) analysis reveals that relative to their neighbors outside the inundation boundary, the group inside the boundary is 7% less likely to work in a startup during the two years after a flood. This effect is pronounced, considering the affected residents experience only several inches of immersion at the peak of the flood (or equivalently about $13,000 in property damage). Given the natural forces behind floods, residents near the inundation line share a common labor market, which is confirmed in various tests. Thus, the above finding likely reflects an impact of the wealth damage on personal entrepreneurial

career choices rather than a change in labor demand.

Although RD provides a sharp identification, the inference is inevitably confined to a narrow support, namely, the 0.1-mile-wide bands around the inundation line in this case. To check the external validity of the result, I next conduct a difference-in-differences (DID) analysis on a broader sample, exploring elevation as a proxy for flood exposure. I find that residents living in elevations falling into the bottom three deciles are 4.5% less likely to work in a startup after the flood relative to their neighbors in the top three deciles within the same county. The analysis lends further support to the hypothesis that suffering flood damage has a deterrent effect on individuals' decision to work in startups.

Such an effect could be attributed to impaired household wealth as well as to the psychological impact of watching one's home being flooded, although the former is more likely to be first-order. To confirm the wealth impact, I conduct a partitioning analysis of homeowners versus renters. Because homeowners disproportionately suffer financial loss from floods, the fact that the effect is concentrated on homeowners favors the wealth channel as opposed to a pure psychological impact. In other words, the lowered propensity to work in startups is likely to be a consequence of wealth loss associated with a flood.

Even after confirming the wealth impact, which of the two forces below are at work is unclear. First, impaired household wealth can reduce an individual's ability to smooth consumption in the event of a negative income shock. This consideration could a priori cause an individual to become averse to working in a startup, especially in an incomplete market where employment risks cannot be perfectly hedged. Second, with nonpecuniary benefits being a potential attraction of many startup jobs, wealth loss may dampen an individual's pursuit of these benefits if they are viewed as "luxury goods." The first hypothesis predicts individuals will refrain from high-risk startup jobs. By contrast, the second implies the popularity of jobs with high nonpecuniary benefits would decline among those affected by floods. I find low-elevation residents particularly refrain from startup jobs involving higher income risks than their previous employment, consistent with the self-insurance mechanism. They do not avoid startup jobs with an income downgrade (and potentially with nonpecuniary-benefits upgrade), inconsistent with the luxury-goods consumption channel.

The career distortions caused by impaired household wealth could have long-term ram-

ifications. The reluctance to take risks in one's career is likely to limit the investment in human capital, and therefore may dampen an individuals' income growth (Shaw, 1996). Indeed, the residents just inside the inundation line in my sample experience a 5% decline in labor income relative to their neighbors outside the inundation line in the six and a half years after the flood.[1] This finding is economically significant: the effect translates to about 62 cents of income loss in response to $1 of property damage.[2]

This study contributes to several strands of literature. First, the findings further our knowledge about the determinants of human capital in startup firms. Previous studies mainly focus on the incentive to found a startup (Hurst and Lusardi, 2004; Adelino, Schoar, and Severino, 2015; Schmalz, Sraer, and Thesmar, 2017; Gottlieb, Townsend, and Xu, 2019), while few studies have explored the incentive to work for startups. Talents are particularly important at the early stage of a firm when the organizational capital has not been formed (Baron et al., 2001; Choi et al., 2021). The paper sheds light on an important determinant of one's decision to work in a startup (i.e., household wealth).

This paper also contributes to an emerging body of literature on the repercussion effect of climate risks. The increasing frequency of natural disasters has been broadly viewed as a realization of climate risks (Nordhaus and Boyer, 2000). Several studies have investigated how natural disaster experience affects firm decisions, such as capital structure decisions and liquidity management (Bernile et al., 2017; Dessaint and Matray, 2017). To my best knowledge, this paper is the first to provide micro-level evidence on how climate risks affect individuals' career decisions.

The remainder of this paper is organized as follows. Section 2 discusses the empirical setting and data. Section 3 details the research design and baseline results. Section 4 sheds light on the economics behind the effect. Section 5 analyzes the long-run impact. Section 6 concludes.

[1]The horizon of the analysis is determined by the time span of the Longitudinal Employer Household Dynamics (LEHD) data coverage.

[2]This estimate can be interpreted as a lower bound of the income-wealth sensitivity because the unobserved partial insurance for flood may lower the financial damage suffered by the inundated group.

2. Data and Overview

2.1. Flood Data

To understand how household wealth affects an individual's decision to work in a startup, I explore natural disasters as a negative shock to household wealth and integrate two key elements of data input: flood data and individual employment information.

Floods are the most devastating climate disaster by frequency and damage. For example, according to the Center for Research on the Epidemiology of Disasters (CRED), floods caused direct financial losses of $285.1 billion in 2017, topping the list of economic losses associated with natural disasters. The frequency and severity of flood events have significantly increased in the last few decades, with about 60% of the counties in the U.S. experiencing at least one significant flood from 2012 to 2016, according to the Federal Emergency Management Agency (FEMA).

The primary direct consequence of a flood is damage to properties and other structures. According to CRED, a major flood can cause tens of billions of dollars in property damage. The damage is likely to have a real impact on households' financial situation because flood risks cannot be fully hedged. Despite the existence of flood insurance, the insured losses typically account for only a fraction of the total property damage suffered in a flood, due to the low take-up rate and the partial nature of the flood insurance policies.[3]

My analysis focuses on five major floods from 2005 to 2010 in five states: Tennessee, Indiana, Iowa, Pennsylvania, and California.[4] These floods were the five most significant ones from 2003 to 2011 in the 17 states where the Longitudinal Employer-Household Dynamics (LEHD) data are accessible.

To identify these floods, I start with the sample of all federal government disaster dec-

[3]Because standard homeowner insurance policies exclude flood damage, the extent of coverage depends on the flood insurance policy purchased separately by homeowners. However, due to the high cost of such plans and limited awareness of potential flood risks, the take-up rate of flood insurance is typically low. For example, according to the estimates provided by Milliman Inc., fewer than 10% of the households affected by Hurricane Florence in 2018 had flood insurance. Even flood insurance plan holders face a limitation on the amount of property damage that can be reimbursed. For instance, federal flood insurance in the U.S. covers no more than $250,000 of damage minus deductibles for a residential property.

[4]The five flood events include the June 2008 Midwest floods in Indiana, 2010 Tennessee floods in Tennessee, June 2008 Midwest floods in Iowa, Los Angeles County flood of 2005 in California, and Mid-Atlantic United States flood of 2006 in Pennsylvania.

larations associated with a flood from 2003 to 2011 in the 17 states where the LEHD data are available.[5] I use Google Trends statistics to identify the most significant floods in each of the 17 states. In particular, I examine the dynamics of Google search frequency with the keyword "flood" combined with a state name and locate the most significant spike in the search frequency in each state. Next, I match each of the spikes with a flood in the CRED database to obtain the dollar amount of property damage caused by each flood. This procedure allows me to restrict the sample further to the five floods that caused property damage of $1 billion or more. To avoid leaving out any significant floods, I manually check each of the visible spikes in the search-frequency history and find none of them pass the property-damage threshold.

I obtain a list of counties affected by the five floods from the county-level disaster declaration database provided by FEMA. My baseline analysis focuses on the 220 counties subject to a federal disaster declaration associated with the five floods.

To gauge an individual's exposure to the property damage in a flood, I use a battery of GIS datasets. To specify the elevation of a residential property, I obtain from the U.S. Geological Survey (USGS) an elevation map with one-third arc-second resolution,[6] the highest-resolution seamless Digital Elevation Model (DEM) dataset with full coverage of the U.S. To explore the micro-level variation of flood damage within a community, I extract high-resolution inundation maps for nine counties of Iowa, Indiana, and Pennsylvania from the Iowa Flood Center and the National Oceanic and Atmospheric Administration (NOAA). The inundation maps can pin down which properties are inundated within a block. Combining the GIS data with individual-level residence location information from the LEHD allows me to track how much the flood affected each individual. I use ArcGIS, software that specializes in the processing of GIS data, to integrate the GIS data and employment data at the individual level.

[5]The period of interest is determined by the availability of LEHD data. In most states, the LEHD data span from 2000 to 2014. To ensure sufficient data are available for both pre- and post-shock periods, I require a disaster to occur at least three years after the starting year in the LEHD sample and three years before the ending year.

[6]Ground spacing is approximately 10 meters north/south, but is variable east/west due to convergence of meridians with latitude.

2.2. Employment Data

To obtain individual employment information, I use the LEHD, a large-scale confidential administrative dataset made available by the U.S. Census Bureau. The LEHD provides a longitudinal match between employer and employee from 2000 to 2014. The core dataset records the income from each job (i.e., an employer-employee combination) in a large panel at the employer-individual-quarter level. Being longitudinally linked at both the individual and firm levels, the data track quarter-to-quarter job transitioning for each individual. The data cover all individuals in the unemployment insurance wage records, which account for more than 90% of civilian wage jobs in the U.S. As mentioned, the batch of LEHD data available to this project covers 17 states, including Arizona, Arkansas, California, the District of Columbia, Delaware, Illinois, Indiana, Iowa, Kansas, Maine, Maryland, Nevada, New Mexico, Oklahoma, Pennsylvania, Tennessee, and Texas.

The LEHD is a good fit for this study for two reasons. First, the firm-age information from the LEHD helps identify the individuals who work for a startup at a given point in time. Following Babina (2019), I define a startup as an employer firm up to two years old and focus on individuals' career choices between a startup and a more mature firm conditional on being employed. Second, among other individual characteristics,[7] the LEHD infrastructure provides the residence location of each person in the form of coordinates to a precision of six decimal places. The residence location information is the key input needed to link the LEHD to the GIS datasets at the individual level, which allows me to specify the extent to which an individual suffers from property damage in a flood.

In addition to employment records, I obtain property values and individuals' homeownership status from Corelogic deeds and tax-roll records, a leading dataset that covers the profiles of more than 147 million residential and commercial properties in the U.S. The data provide detailed information about property transactions, property characteristics, and mortgage terms and performance. The key variables include transaction price, owner's name, mortgage loan amount, loan term, appraisal price, property location, and so on.

I merge the Corelogic data and the LEHD at the individual level based on the individ-

[7]The LEHD also provides individual-level demographic attributes, such as gender, age, and educational attainment.

uals' name and physical address, which has not been done in prior research. The match is accomplished using the Person Identification Validation System (PVS) from the U.S. Census Bureau. This data integration represents one of the earliest efforts to construct a comprehensive dataset that provides both detailed homeownership information and employment dynamics at the individual level with broad coverage of the U.S. population.

2.3. Sample Overview and Summary Statistics

The analysis in this paper relies on two main samples. The broader sample used in the DID identification strategy consists of individuals in the LEHD data residing in one of the 220 counties in which the federal government declared a disaster associated with one of the five floods specified above, and those that live in a property with an elevation in the bottom or top three deciles within a county. Because the sample satisfying the above criteria is too massive to be assessed within a reasonable time frame, for the baseline DID analysis, I use a 2% randomized subsample based on the ending digits of the unique individual identifier in the LEHD following the convention of the LEHD literature (e.g., Herkenhoff, Phillips, and Cohen-Cole (2016)). This method yields a sample of around 218,000 individuals, with around 109,000 living in a property with an elevation in the bottom three deciles, and around 109,000 living in the top three deciles.[8] The analysis builds on two years before and two years after each flood in a staggered setting.

Panel A of Table I reports the summary statistics of the sample. *Elevation* is the elevation of an individual's residential property in feet. $D^{Startup}$ denotes a dummy variable equal to 1 if in a given quarter, an individual's primary job is working for a startup up to two years old, and 0 if he/she works for a more mature firm. *Education* is the highest degree of a person based on a 4-point scale (1 = *less than high school* and 4 = *a bachelor's or advanced degree*). *Age* is the age of an individual at the time of the flood. *Male* denotes a dummy variable equal to 1 for a male, and 0 for a female. *Income* is an individual's average quarterly salary from the primary job over the sample period, where a primary job is defined as the job that provides the highest income if an individual has more than one job in the same

[8]This study only reports the rounding numbers of observations for tests involving Census data, following the disclosure requirement of the U.S. Census Bureau.

quarter. Columns (1)–(3) report the mean, standard deviation, and pseudo median[9] of the above variables, respectively. Around 11% of the people in my sample work for a startup in a given quarter conditional on being employed, close to the startup employment rate in private-sector establishments according to the U.S. Census Bureau (around 11.6% during the sample period). An average person in my sample is a 38 years old, holds some college or associate degree, and earns about $7,800 per quarter, comparable to the statistics reported by the U.S. Census Bureau.

To explore the micro variation in inundation, the RD analysis builds on a sample of 6,700 individuals living in the nine counties where the inundation maps are available in Iowa, Indiana, and Pennsylvania, and who resides in a property within 0.1-mile-wide bands on both sides of the actual inundation boundary. Unlike the DID sample, the RD sample includes all individuals in the LEHD (rather than a randomized sample) who meet the above criteria to enhance the power of the test. Within the sample, about 2,500 lived within the inundated area at the time of the flood, and about 4,200 lived outside the flooded area. Similarly, the RD analysis builds on two years before and two years after each flood in a staggered setting.

Panel B of Table I reports the statistics of the RD sample. The comparison between Panels A and B shows the RD sample is representative of the broader population despite the focus on the narrow bands around the actual flood line. The RD sample has comparable attributes to the broader sample reported in Panel A, except for exhibiting a slightly lower quarterly income level than the DID sample.

3. Household Wealth and Career Choices

This section presents the empirical strategies and reports the results, with the goal of understanding how household wealth loss affects individuals' decisions to work in startups. Using floods as a negative shock to household wealth, I conduct two empirical strategies to investigate how household wealth affects entrepreneurial career choices. In an RD framework,

[9]The U.S. Census Bureau does not allow disclosure of the median of any variable. To accommodate the disclosure requirement, I report the pseudo median, which is defined as the mean of all observations between the 49th and 51st percentiles.

the first strategy exploits the spatial discontinuity in property damage around the boundary of the areas actually inundated in a flood. I compare the residents living within 0.1-mile-wide bands on both sides of the boundary of inundated areas regarding their entrepreneurial career choices around a flood. To provide external validity, I further compare low- and high-elevation residents within the same county regarding their pre-post change in their decisions to work in startups. This section elaborates on the two empirical strategies and reports the results.

3.1. Regression Discontinuity Design

To understand how household wealth affects individuals' entrepreneurial career choices, a natural starting point is to compare individuals with heterogeneous levels of wealth. However, such a simple reduced-form analysis is likely to yield a biased estimate due to the endogeneity of wealth.[10] To address this issue, I use a flood as an exogenous shock to household wealth. Comparing the job choices of the same person before versus after the shock controls for the impact of unobserved heterogeneity across individuals. Also, the randomness in a flood's location provides a natural control group that can serve as a proxy for the counterfactual state in the absence of wealth damage.

Accordingly, I employ an RD design around the boundary of *the actual inundated areas* in a flood. The exact location of the flood frontier is affected by the place, intensity, and duration of precipitation, which makes the inundation line inherently unpredictable. Therefore, within a local area where residents face similar ex-ante flood risks, self-selecting into or out of the actual flooded houses is extremely hard, if not impossible, even if people want to do so before the shock. My analysis builds on such a local area. In particular, I compare the residents living within 0.1-mile-wide bands inside (treated group) and outside the inundation boundary (control group), in terms of their decision to work in startups two years before versus two years after the flood. As expected, the treated and control groups are ex-ante indistinguishable, lending support to the causal interpretation of the analysis.

Another advantage of the setting is that it disentangles the effect on individuals' desire

[10]For example, wealthy people might tend to have more diverse skill sets and experiences, which makes them fit for startup jobs regardless of their wealth (Lazear, 2004).

to work in startups from the effect on the labor demand of their employers and other firms. As demonstrated in the covariate balance analysis, the employers of the treated and control groups are inseparable in terms of their exposure to flood, which is a natural outcome of the proximity between the two groups of individuals (typically one or two blocks away from each other) and indicates that the two groups share a common pool of job opportunities.

Covariate Balance Analysis

Before discussing the RD regressions in more detail, I test the identification assumptions of the strategy. The causal interpretation of the analysis depends on the premise that the residents in the narrow bands on both sides of the boundary are balanced in their predetermined covariates. Although the balance of unobservable covariates is inherently non-testable, I shed light on this premise through observed characteristics.

More specifically, I compare a battery of predetermined attributes for residents living in the 0.1-mile-wide bands on both sides of the inundation boundary. Table II reports the result, where $D_{Pre}^{Startup}$ denotes a dummy variable equal to 1 if an individual works in a startup in the quarter before the shock, and 0 if he/she works for a more established firm. *Income* is an individual's average quarterly income in the two years before the shock, and *Home Value* is the nominal home value in U.S. dollars at the time of the flood. *Home Flood Risk* (*Employer Flood Risk*) captures the ex-ante flood risk exposure of an individual's residence location (workplace). In particular, the flood risk is defined as a categorical variable on a scale of 0–2 (2 (1) = *an address is within a 100-year (500-year) pre-designated flood zone and 0 = not in a flood zone*). *Employer Inundation Status* is a dummy variable set to 1 if in the quarter before the shock, a person worked in an establishment that would be inundated in the flood, and 0 if he/she worked outside the to-be-flooded area. *Age*, *Male*, and *Education* are defined in Table I.

The treated and control groups do not exhibit significant discrepancy in any of the predetermined attributes, lending support to covariate balance. In particular, the two groups are indistinguishable in their demographic profiles, such as age and gender. They are also comparable in proxies for socioeconomic status, including education, income, and home value. More interestingly, the treated and control groups face similar ex-ante flood risks, as

measured by the designated flood zones of their houses on the flood plane. This observation suggests the excessive flood damage the treated group suffered in comparison with the control group is unlikely to be caused by an endogenous choice to live in an area with a higher flood risk. Overall, the balance of the covariates is consistent with the assumption that the assignment of the treatment between the two groups is quasi-random.

One may still be concerned that even if the treated and control individuals are indistinguishable, their (potential) employers may be differently affected by the flood. In this case, the observed divergence between the two groups' entrepreneurial career choices may be explained by the employers' hiring/layoff decisions rather than their desire to work in startups.

This concern is alleviated by the observation that the two groups' employers are equally likely to reside in the inundated area (see line 9 of Table II). The finding suggests the labor market is not segmented around the flood boundary and the two groups' employers are similarly affected by the flood. Line 10 of Table II further shows their employers are exposed to the same level of flood risks ex-ante, reinforcing the assumption that the two groups of individuals face a common labor market.

Besides the covariate balance, the RD design also assumes individuals cannot perfectly manipulate their residence location around the boundary in anticipation of the flood. Given the 0.1-mile-wide bands on both sides of the boundary, manipulation requires precise prediction of the flood frontier. Because the inundation boundary is primarily determined by the location, intensity, and duration of precipitation, such a prediction is understandably beyond the capacity of an average resident. In this sense, the perfect manipulation is reliably unavailable in this setting. A McCrary density test further supports the absence of perfect manipulation: no significant discontinuity in the population density function presents at the boundary of the inundation area.[11]

[11]Notably, using the RD design in a DID framework requires less strict identification assumptions than what I have tested. Even if the treated and control groups around the boundary of the inundated areas differed significantly in some attributes, the identification strategy would remain valid as long as the discrepancy did not drive a dispersion in employment decisions concurrently with the flood. In other words, the above findings in line with the strict version of the identification assumptions lend strong support to the validity of the research design.

Regression Analysis

Having verified the identification assumptions, I conduct a regression analysis that compares the residents in the 0.1-mile-wide bands on both sides of the inundation boundary regarding their decisions to work in startups around the flood. In particular, I estimate a DID specification of the following form at the individual-quarter level:

$$D_{s,i,t}^{Startup} = \alpha_{s,i} + \alpha_{s,t} + \beta_1 D_{s,t}^{Post} \times D_{s,i}^{Inundated} + D_{s,t}^{Post} \times (\sum_{k=1}^{K} \gamma_k D_{s,i}^{k}) + \varepsilon_{s,i,t}. \quad (1)$$

The regression builds on "stacked" panel data that covers 6,700 individuals in nine counties of Pennsylvania, Iowa, and Indiana over the four years around each flood. s, i, k, and t denote a flood event, an individual, a county, and a quarter, respectively. $D_{s,t}^{Post}$ is a dummy variable equal to 1 if quarter t is within two years after flood s (including the quarter of the flood), and 0 if quarter t is within two years before the flood. $D_{s,i}^{Inundated}$ is a dummy variable equal to 1 if individual i lived inside the inundated zone before flood s, and 0 otherwise. The outcome variable of interest $D_{s,i,t}^{Startup}$ is a dummy variable set to 1 if individual i works for a startup during quarter t, and 0 if he/she works for an established firm. As mentioned above, I define a startup as a firm up to two years old. Based on the definition of the outcome variable, all tests hereafter will condition on an individual being employed in a quarter unless otherwise specified.

To control for individual heterogeneity and macro-level trends, the above specification incorporates individual fixed effects $\alpha_{s,i}$ and quarter fixed effects $\alpha_{s,t}$. The interaction terms of $D_{s,t}^{Post}$ and the full set of county fixed effects further ensure the comparison is conducted between the residents around the same flood line. In particular, $D_{s,i}^{k}$ is a dummy variable set to 1 if a person lives in county k at the time of the flood, and 0 otherwise. The key coefficient β_1 captures how much more likely an inundated resident is to work in a startup after the shock, relative to a neighbor whose home is marginally unflooded. I expect β_1 to be significantly negative if experiencing wealth loss in a flood discourages people from working in startups.

Table III reports the result. The significantly negative coefficient on $D^{Post} * D^{Inundated}$ suggests after a flood, the residents in inundated areas are less likely to work in a startup,

relative to their neighbors residing just outside the flooded area. Considering the treated and control groups are indistinguishable on pre-determined attributes, the result suggests experiencing flood damage has a negative, causal effect on the propensity to pursue entrepreneurial careers.

In terms of the magnitude, the coefficient in column (1) translates into a 7% decline in the likelihood of working for a startup, relative to the baseline rate of working in a startup.[12] Interestingly, this effect is a consequence of a relatively modest flood damage: on average, the treated group experiences eight inches of water in their houses at the peak of the flood, or equivalently about $13,000 in property damage. In other words, the analysis suggests a wealth loss of $13,000 can have a pronounced deterrent effect on individuals' decision to work in startups.

Graphical Evidence

The above analysis shows residents in the narrow band inside the inudation boundary become less likely to work in startups after the flood. To further understand how the effect is distributed, I visualize the change in the likelihood of pursuing startup jobs across the distances from the boundary in Figure 1. Subfigure (a) reports the inundation depth around the boundary, where the inundation depth is defined as the depth of water above ground at the peak of the flood. The horizontal axis is the distance between a residential location and the boundary, which is set to negative outside and positive inside. Each point represents the average inundation depth for all residential properties within a 0.005-mile-wide band. Because the slope around the boundary is mostly gentle, we would expect a kink rather than a jump in inundation depth around the boundary. Indeed, the inundation depth remains 0 outside the boundary and gradually increases inside the boundary. Because inundation depth is highly correlated with flood damage (Dutta, Herath, and Musiake, 2003), the above observation suggests residents in the inner flooded area experience greater property damage.

How do individuals' responses vary with the distance from the boundary? Subfigure (b) reports the change in their entrepreneurial career choices around the shock. Each point

[12] I estimate the magnitude by dividing the coefficient in column (1) (-0.780) by the likelihood of working in startups in Table I.

represents the pre-post change in the likelihood of working for a startup for an average person residing within a 0.005-mile-wide band. Likewise, the presence of a kink at the origin is consistent with the absence of physical barriers around the flood frontier. While the propensity to work for startups is mostly unchanged outside the boundary, the residents inside become less likely to pursue startup jobs after a flood. In particular, the deterrent effect increases with the distance to the boundary (and therefore the inundation depth), lending support to a causal relationship between flood damage and individuals' entrepreneurial career choices.

The presence of kinks at the boundary indicates a regression kink (RK) design is also applicable to this setting. Accordingly, I report the RK analysis in Appendix A.1, which confirms the robustness of the RD findings.

Further Discussion about the RD Analysis

The above RD results are consistent with an dampening effect of household wealth loss on individuals' entrepreneurial career choices. A potential concern is the above findings could be driven by the differential labor demand the treated and control groups face after the flood. To address the concern, I have shown the employers of the two groups at the time of the flood are exposed to similar flood risks and are equally likely to be inundated in the flood. To further verify that the hiring demand and layoff decisions of the two groups' employers do not diverge after the flood, I examine the two groups' unemployment rates around the flood and report the results in Panel B of Table III. The outcome variable $D^{unemployment}$ is a dummy variable equal to 1 if an individual does not have a paid job in a given quarter, and 0 otherwise. Likewise, the analysis builds on the individuals who live within 0.1 miles of the inundation boundary at the time of a flood and who are employed in at least one quarter during the two years before the shock.

The first two columns report the difference between the treated and control groups in their unemployment rates before and after the shock, respectively. The coefficient estimates show the two groups do not exhibit significant differences in their likelihood of being unemployed in either periods. Columns (3) and (4) further confirm the observation: the insignificant coefficients on $D_t^{Post} * D_i^{Inundated}$ show the two groups do not diverge in their propensity of

unemployment after the shock. Overall, the above evidence suggests the RD effect is more likely to be attributed to personal career choices than a change in labor demand.

3.2. Difference-in-Differences Design across Elevations

So far, I have explored an RD setting to identify the causal effect of household wealth on individuals' entrepreneurial career choices, using granular geographic variations in inundation depth around the flood frontier. The analysis is informative about the causal relationship but comes with two potential caveats. First, because the analysis builds on a local area with a relatively small population, the power of the test could limit my ability to conduct further partitioning analyses to understand the mechanisms behind the effect. Second, the inference is inevitably confined to a narrow support (the 0.1-mile-wide bands around the flood line in this case), which makes the external validity an empirical question.

Accordingly, I complement the RD tests with a broader-sample analysis that covers most residents in the flooded counties within the five states specified above. In a DID framework, I compare the low- and high-elevation residents before versus after a flood, using elevation as a proxy for flood damage. The DID design serves two purposes. First, the analysis helps verify the external validity of the RD results. Second, the enhanced power of the test enables me to conduct a variety of partitioning analyses to shed light on the economics behind the effect.

First-Stage Analysis: Elevation and Financial Damage

A key premise of the analysis across elevations is that the low-elevation residents suffer more financial damage from floods than high-elevation residents. To test this assumption, I conduct a set of first-stage analyses in this subsection. Because the outcomes of interest concern the financial damage to a property rather than the employment decision of an individual, the analysis builds on samples that consist of 35,609–743,533 properties from Corelogic data, depending on the data availability of outcome variables in each specification.

In particular, I start by running the following regression:

$$Outcome_{s,i} = \alpha_s + \beta_1 Elevation_{s,i} + \sum_{k=1}^{K} \gamma_k D_{s,i}^k + \varepsilon_{s,i}, \tag{2}$$

where i, k, and s denote a property, a county, and a flood event, respectively. $Elevation_{s,i}$ is defined as the elevation of a given property. Two outcomes of interest emerge: $InunDepth_{s,i}$ and $Damage_{s,i}$. $InunDepth_{s,i}$ is the amount of water above ground at the location of property i during flood s. $Damage_{s,i}$ denotes the flood damage to property i on a 5-point scale (1 = *no damage* and 5 = *total destruction*). Because flood damage is not observable for most of the main sample, the test builds on an out-of-sample flood event in Texas in 2017, of which the damage data are provided by NOAA for each affected property. The key coefficient β_1 captures how much the elevation determines property damage/inundation in a flood.

Columns (1)–(4) of Panel A in Table IV present the results. Columns (1) and (2) report the effect on $InunDepth$ and columns (3) and (4) examine $Damage$. The specifications in columns (1) and (3) (columns (2) and (4)) incorporate county fixed effects (zip code fixed effects). The significantly negative coefficients on $Elevation$ suggest lower-elevated properties are more likely to be inundated and damaged due to a flood. Regarding the magnitude, a one-standard-deviation decrease in elevation corresponds to a 0.2-standard-deviation increase in property damage.

To test a flood's impact on home values more directly, I run the following regression at the property-year level:

$$HomeValue_{s,i,t} = \alpha_{s,i} + \alpha_{s,t} + \beta_1 D_{s,t}^{Post} \times Elevation_{s,i} + D_{s,t}^{Post} \times (\sum_{k=1}^{K} \gamma_k D_{s,i}^k) + \varepsilon_{s,i,t}, \tag{3}$$

where $HomeValue_{s,i,t}$ is the value of property i in year t measured by the latest assessed pricing of the property. Similar to equation (1), the specification includes individual fixed effects $\alpha_{s,i}$, year fixed effects $\alpha_{s,t}$, and the interaction terms between $D_{s,t}^{Post}$ and county fixed effects $D_{s,i}^k$. Because, among the five states of interest in the DID sample, Tennessee is the only state where assessed pricing is available for both the pre- and post-flood periods, the

estimate builds on the subsample in Tennessee. I focus on one year before and one year after the shock to capture the immediate change in appraisal value around the flood.

The coefficient β_1 represents a flood's impact on the value of high-elevation properties relative to low-elevation properties in the same county. I expect a negative estimate of β_1 if a flood leads to a disproportionate impairment of low-elevation houses.

Columns (5) and (6) of Panel A in Table IV report the results. Column (5) focuses on the dollar value of a property (in thousands) measured by its appraisal value, and column (6) examines the housing value denominated by the last appraisal price in the pre-shock period.[13] Regardless of the housing-value measures, the coefficients on $D^{Post} \times Elevation$ remain significantly positive, consistent with a negative association between elevation and home-value loss in a flood.

Overall, low-elevation properties are more prone to inundation and experience greater value loss than high-elevation houses in the same county. These findings confirm elevation as a legitimate proxy for flood exposure.

Reduced-Form Analysis: Elevation and Entrepreneurial Career Choices

Based on this premise, I conduct a DID analysis to compare the change in entrepreneurial career choices for individuals residing in low- and high-elevation areas around the flood. In particular, I run an individual-quarter-level regression of the following form:

$$D_{s,i,t}^{Startup} = \alpha_{s,i} + \alpha_{s,t} + \beta_1 D_{s,t}^{Post} \times D_{s,i}^{LowElev} + D_{s,t}^{Post} \times (\sum_{k=1}^{K} \gamma_k D_{s,i}^{k}) + D_{s,t}^{Post} \times (\sum_{j=1}^{J} \tau_j D_{s,i,t-1}^{j}) + \varepsilon_{s,i,t}. \tag{4}$$

The regression builds on the "stacked" panel data that covers the relevant individuals for each of the five flood events over the four years around each flood. s, i, j, k, and t denote a flood event, an individual, a firm (an employer), a county, and a quarter, respectively. I focus on the bottom and top three deciles by elevation to boost the power of the test.[14]

[13] Because some homeowners may choose to repair their home immediately after the flood and before the next appraisal, the assessed pricing may include the value of repairs, and therefore underestimates the damage arising from a flood.

[14] The baseline results are robust to the use of alternative thresholds such as the top and bottom quintiles, according to untabulated robustness tests.

Accordingly, $D_{s,i}^{LowElev}$ is a dummy variable equal to 1 (0) if the elevation of the property where individual i lived at the time of the flood falls in the bottom (top) three deciles within a county. Similar to equation (1), the specification includes individual fixed effects $\alpha_{s,i}$, quarter fixed effects $\alpha_{s,t}$, and $D_{s,t}^{Post} \times$ county fixed effects.

One may be concerned that the high- and low-elevation residents might work for different types of firms before the flood, and the differential impact of floods on their employers, rather than their personal career choices, may in turn explain the post-shock divergence between the two groups. Although the covariate balance of the RD analysis suggests the effect is not driven by a flood's differential impact on employers, I further confirm whether the same assumption holds for the DID setting, by incorporating the interaction terms of $D_{s,t}^{Post}$ and the full set of firm fixed effects $D_{s,i,t-1}^{j}$ in the specification. In particular, $D_{s,i,t-1}^{j}$ is a dummy variable equal to 1 if individual i works for employer j in quarter $t-1$, and 0 otherwise. Using the firm $\times$ post fixed effects controls for the flood's impact on employers and would attribute the effect, if any, to individual career choices rather employers' hiring demand/layoff decisions.

Essentially, the strictest form of the analysis compares the pre-post change in career choices for two individuals living in the same county at the time of the flood and *working for the same firm in the previous quarter*, one of whom resides in a low-elevation area and the other in a high-elevation area. If flood damage discourages people from working in startups, I expect a negative estimate of β_1.

Panel B of Table IV reports the estimation based on two years before and two years after the staggered flood events. The coefficient on $D^{Post} \times D^{LowElev}$ loads significantly negative regardless of the specifications, suggesting low-elevation residents experience a greater decline in their likelihood of working for startups. Given the excessive exposure to financial damage among low-elevation residents, this finding is consistent with the dampening effect of household wealth loss on individuals' entrepreneurial career choices. The magnitude of the effect is also comparable to the RD findings. Relative to the baseline rate of startup employment, an individual living in a low-elevation area becomes 4.3%[15] less likely to work

[15]I estimate the magnitude by dividing the coefficient in column (1) (-0.487) by the probability of working in startups from Table I.

in a startup than a former colleague living in a high-elevation neighborhood in the same county. Overall, the above results suggest the relationship between financial damage and entrepreneurial career choices carries beyond the local area around the flood frontier, extending the external validity of the RD findings.

Parallel-Trends Analysis

The causal interpretation of the above DID findings relies on the assumption that the entrepreneurial career choices of high- and low-elevation residents would have experienced similar trends around the shock absent the floods. Although the parallel trends in the counterfactual state without a flood are inherently non-testable, I shed light on the assumption by examining pre-existing trends. More specifically, I test how the decision to work in startups evolves around the shock for the high- and low-elevation residents by estimating the regression

$$D_{s,i,t}^{Startup} = \alpha_{s,i} + \alpha_{s,t} + \sum_{\tau=-8}^{7} \beta_\tau D_{s,t}^{\tau} \times D_{s,i}^{LowElev} + D_{s,t}^{Post} \times (\sum_{k=1}^{K} \gamma_k D_{s,i}^{k}) + \varepsilon_{s,i,t}, \qquad (5)$$

where $D_{s,t}^{\tau}$ equals 1 if quarter t is the τth ($-\tau^{th}$) quarter following (prior to) flood s, and 0 otherwise. The coefficient of interest β_τ measures the gap between the two groups in terms of their likelihood of working in a startup during the τth ($-\tau^{th}$) quarter after (before) the shock.

Figure 2 reports the results. The horizontal axis represents the quarter relative to the flood events, and the vertical axis represents the estimated β_τ for the corresponding quarter. No significant differences emerge between the low- and high-elevation residents for all quarters before the shock, which is consistent with the parallel trends. Also, the graph confirms the differential impact of financial damage in flood on low- and high-elevation individuals. In each quarter after the flood, low-elevation residents are significantly less likely to work in startups relative to high-elevation neighbors in the same county at the 5% confidence level. Note the magnitude of the effect is overall progressive during the two years after the shock. This observation suggests over time, flood experience causes an increasing number of people to refrain from startup jobs, lending support to a persistent effect of adverse wealth shocks

on individuals' career choices.

Additional Discussion about the DID Analysis

The above analysis verifies the parallel trend assumption and alleviates the concern about potential confounders. Due to the random nature of when and where a flood happens, a confounder that coincides with each of the staggered floods is very unlikely to exist. To further confirm the divergence between the two groups is not driven by a confounder, I conduct placebo tests on the neighboring counties unaffected by the flood. As detailed in Appendix A.2, the result is consistent with the absence of a confounder and strengthens the causal inference.

How should we interpret the change in entrepreneurial career choices? The lowered likelihood of working in startups among those living in low-elevation areas can be attributed to two factors: (1) the reduced transitioning from mature firms into startups, and (2) the lowered retention rate within startups. My analysis suggests both forces contribute to the effect. As reported in Appendix A.3, the lowered likelihood of joining a startup from an established firm accounts for about 70.5% of the baseline effect, and the reduced retention rate within young firms explains 29.5%. Thus, a typical example of the effect is as follows. After her house had been flooded, an individual who was working for a big firm was less likely to join a newly established startup than her coworker who lived in the same county but had not experienced flood damage.

4. Mechanism Analysis

So far, the empirical analysis shows that experiencing flood damage is likely to discourage people from working in a startup. This section further explores the economics behind this phenomenon.

4.1. Wealth Effect versus Psychological Impact

A natural question is whether the divergent responses of the residents with differential flood exposure indeed reflect a household wealth effect. As discussed above, property damage

accounts for the vast majority of the direct cost associated with a flood, which can impair household wealth through the lowered home value or additional expense on home repair. Because the treated and control groups are presumably subject to a similar change in local economic conditions, the effect identified above is likely to be attributable to the excess flood damage borne by the treated group relative to the control group.

Alternatively, the phenomenon could be explained by a psychological impact. A person may be traumatized by disaster experience, and thus becomes less willing to venture in his/her career. Although the residents inside and outside the flood line both observe the disaster event in the neighborhood, the treated group could still experience a greater mental shock due to the inundation of their own dwelling places, and therefore becomes less willing to work in startups.

To test whether the above finding reflects the wealth effect or a pure psychological impact, I explore the cross-sectional variation in homeownership across individuals. Because homeowners suffer more financial damage than renters in a flood, I expect a stronger impact on homeowners if household wealth loss is a main explanation of the effect. By contrast, since both groups experience floods in their homes, I would not expect a strong differential effect if psychological impacts prevail.

Accordingly, I conduct a triple-differences analysis comparing the effect on homeowners versus non-homeowners. The analysis builds on the 2% randomized sample used in the DID analysis above, complemented by a 10% randomized sample of homeowners identified from the merger between the LEHD and Corelogic data. The reason I use the expanded sample is to compensate the loss of variation due to the partial matching between Corelogic and the LEHD.[16]

To control for potential self-selection into homeownership in anticipation of a flood, I focus on the homeowners who were living in their properties at the time of the flood and became homeowners at least one year before the shock. In particular, I run an individual-

[16]In particular, the matching identified less than 10% of the individuals as homeowners in the DID sample, which limits the power to detect the differential effect on the homeowners versus non-homeowners (i.e., a mix of renters and unidentified homeowners). The expanded sample helps restore the testing power and makes a triple-differences analysis feasible.

quarter-level regression of the following form:

$$D_{s,i,t}^{Startup} = \alpha_{s,i} + \alpha_{s,t} + \beta_1 D_{s,t}^{Post} \times D_{s,i}^{LowElev} \times D_{s,i}^{Homeowner} + \beta_2 D_{s,t}^{Post} \times D_{s,i}^{LowElev} + \beta_3 D_{s,t}^{Post} \times D_{s,i}^{Homeowner} + D_{s,t}^{Post} \times (\sum_{k=1}^{K} \gamma_k D_{s,i}^{k}) + \varepsilon_{s,i,t}, \quad (6)$$

where $D_{s,i}^{Homeowner}$ is a dummy variable that indicates whether an individual is identified as a homeowner from the matching between Corelogic and the LEHD. In particular, $D_{s,i}^{Homeowner}$ is set to 1 if individual i lived in his/her own property at the time of flood s and became a homeowner at least one year before the flood, and 0 otherwise. Due to the partial matching between the two datasets, the group with $D_{s,i}^{Homeowner} = 0$ is a mix of renters and unidentified homeowners. Because homeowners are disproportionately exposed to financial damage, I would expect a negative estimate of β_1 if the change in entrepreneurial career choices are induced by the wealth loss in a flood.

Table V reports the result of the triple-differences analysis. The coefficients on $D^{Post} * D^{LowElev} * D^{Homeowner}$ are significantly negative at the 5% level regardless of the specifications. The result suggests homeowners refrain from working in startups more than non-homeowners (i.e., a mix of renters and unidentified homeowners). In particular, the estimated differential effect in column (1) is 70% of the baseline effect in Table IV. If we take this result seriously and further consider the homeownership rate in the areas of interest (60.5%), the estimate suggests 90% of the effect in Table IV can be attributed to homeowners refraining from working in startups. As discussed above, this finding is more consistent with the wealth effect than a purely psychological impact.

Admittedly, experiencing financial damage per se may incur psychological pain, which is likely to affect homeowners more than renters. By the above analysis, I do not intend to differentiate the wealth effect from the psychological effect perfectly correlated with wealth, because the two effects are technically inseparable. The triple-differences analysis is designed to test the household wealth effect versus the component of the psychological effect that is orthogonal to wealth loss, such as the mental effect caused by one's dwelling being flooded regardless of whether he/she needs to pay for the repair.

Another potential explanation for the excess effect on homeowners might be the time they

spend on repairing their home and other tasks. Such distractions may limit their potential effort to search for a new job and makes them less likely to join newly established startups. In this case, the effect is likely to be short-lived because the repair work typically takes no more than several weeks (FEMA, 2010). However, the data show the effect is rather persistent. As shown in Figure 2, the impact on entrepreneurial career choices is progressive in the two years after the shock, which is inconsistent with the distraction hypothesis. Also, home repair work is unlikely to explain the increased likelihood of joining mature firms from startups among the treated group (see Appendix A.3). If the distraction effect were the main mechanism, we would expect to see instead a decrease in the time-consuming transitioning from startups into established firms. Based on the above observation, the concentration of the effect on homeowners is more likley to be attributed to household wealth loss than the post-disaster distraction.

4.2. Why Does Wealth Loss Affect Entrepreneurial Career Choices?

The findings thus far support the hypothesis that household wealth loss discourages individuals from working in startups. However, why such a link would exist is unclear.

The lowered desire to work in startups might be explained by the reduced willingness to take career risks as a result of wealth loss. According to a lifetime consumption model, an important saving motive is to smooth consumption in the event of a negative income shock, based on the assumption that income risks cannot be perfectly hedged through capital markets (e.g., Browning and Lusardi, 1996). The framework suggests a negative shock to household savings can impair individuals' self-insurance capacity, and therefore lower the amount of career risks they are willing to take. Given the negative association between firm age and income risks (Kerr, Kerr, and Nanda, 2015), the above findings can be attributed to the lowered self-insurance capacity, which makes the affected individuals refrain from risky jobs in startups.

Alternatively, the effect might be related to the nonpecuniary benefits of working in startups. A line of research shows pursuing entrepreneurial activities can bring additional benefits on top of financial reward, such as flexibility, variety in work experience, and a sense of independence and control (Moskowitz and Vissing-Jørgensen, 2002; Hamilton, 2000; Hurst

and Pugsley, 2011; Åstebro and Thompson, 2011). If we view such nonpecuniary benefits as luxury goods (Hurst and Lusardi, 2004), impaired household wealth may prompt individuals to cut their consumption of these benefits.

The two candidate mechanisms yield different predictions on which type of startup jobs people will avoid more. If the self-insurance mechanism prevails, the reduced risk-taking capacity would discourage people from pursuing startup jobs with higher income risks than their current positions. Otherwise, the nonpecuniary-benefit mechanism would predict a lowered likelihood of taking startup jobs that provide an upgrade in nonpecuniary benefits.

Although testing the first prediction is apparently feasible, testing the second is more challenging given the unobservable nature of a job's nonpecuniary benefits. To address this issue, I use the income reduction as a result of joining a startup as a proxy for the increase in nonpecuniary benefits associated with the job change, following the literature that interprets the unexplained return gap between regular jobs and entrepreneurial activities as nonpecuniary benefits (Moskowitz and Vissing-Jørgensen, 2002; Hamilton, 2000). This measure builds on the assumption that an unobserved upgrade in nonpecuniary benefits contribute to one's willingness to accept a lower-paying job, which is likely to hold in the context of entrepreneurial activities.

Accordingly, I explore the income levels and risks of different startup jobs to shed light on the two potential mechanisms. Panel A (Panel B) of Table VI reports the partitioning tests on how impaired household wealth affects the decisions to take high-income/low-income (high-risk/low-risk) startup jobs relative to the previous employment. In columns (1) and (2) (columns (3) and (4)) of Panel A, $D^{Startup}_{LowerIncome}$ ($D^{Startup}_{HighIncome}$) is a dummy variable equal to 1 if an individual's startup job in a given quarter provides lower (higher) income than his/her previous employment, and 0 otherwise, conditional on being employed. Columns (1) and (2) (columns (3) and (4)) of Panel B examine $D^{Startup}_{LowerRisk}$ ($D^{Startup}_{HigherRisk}$), a dummy variable equal to 1 if one's current startup job has a lower (higher) income volatility than his/her previous job, and 0 otherwise.[17]

Panel A shows that relative to high-elevation neighbors in the same county, low-elevation

[17]The result is robust to using the residual income volatility of a job that is not explained by the return of observable worker characteristics as the job-risk measure following Gourinchas and Parker (2002), Meghir and Pistaferri (2004), and Krishna and Senses (2014).

residents become less likely to pursue upgrade startup jobs in terms of income. At the same time, the two groups do not diverge in their propensity to take downgrade startup jobs. Because the pursuit of nonpecuniary benefits often comes at the expense of the income, the decreased consumption of such benefits is not very likely to be a main explanation for the lowered desire to work in startups.

Panel B suggests household wealth loss discourages individuals from taking startup jobs with higher income risks than their previous jobs, while their propensity to take low-risk startup jobs is not significantly affected. This finding relates the wealth impact on entrepreneurial career choices to the risky nature of startup jobs. Overall, the concentration of the effect on high-risk, high-income startup jobs is more consistent with the self-insurance mechanism than the luxury-goods-consumption channel.

To further verify whether self-insurance is the underlying mechanism, I investigate the effect of household wealth on risk-taking employment decisions more broadly. In particular, I examine the decisions to work in industries with high employment risks (regardless of whether the employer is a startup or not). If the above effect were related to lowered self-insurance capacity, I would expect individuals to become less likely to work in industries with high employment risks as well.

The result is reported in Panel C of Table VI. The outcome variable, $IndustryRisk_{i,t}$, measures the employment risks of the industry where individual i works in quarter t. Because job turnovers are associated with forced separations, I measure the predetermined employment risk of an industry using the average quarterly separation rate in the five years before the sample period. Following the literature convention, I define the separation rate as the number of separations in a SIC six-digit industry divided by the average employment at the beginning and the end of the quarter (Antoni, Maug, and Obernberger, 2019). The significant coefficients on $D^{Post} \times D^{LowElev}$ in Panel C shows that after a flood, the low-elevation residents become less likely to work in high-risk industries relative to their high-elevation neighbors in the same county, which is consistent with the self-insurance mechanism.[18]

[18]In an average quarter before the shock, the increase in average industry employment risks induced by individuals' transition into higher-risk industries is 1.3%. Relative to this baseline rate, the effect amounts to a 5.6% decrease in the propensity to transition into high-risk industries according to column (1) of Panel C, Table VI.

Another potential explanation for the the dampened desire to pursue startup jobs is the "lock-in" effect of heightened leverage that potentially limits the labor mobility (Brown and Matsa, 2019; Gopalan, Hamilton, Kalda, and Sovich, 2019). In this particular setting, several observations suggest this channel is not the main explanation of the findings. First, the treated group is slightly more likely to move after their home is flooded, which suggests the reluctance to work in startups is not related to the inability to relocate.[19] Second, as shown in Appendix A.3, the transitioning rate into startups and the retention rate within startups both decrease among the affected individuals, which indicates the result is not purely driven by the "lock-in" effect.

Overall, the above observations suggest the lowered self-insurance is likely to be the underlying mechanism through which household wealth affects individuals' entrepreneurial career choices.

5. Long-Run Impact on Personal Income

So far, I have shown evidence that wealth loss discourages individuals from working for startups through impairing self-insurance. The wealth loss is also likely to make individuals refrain from other risky employment decisions such as working in high-risk industries. The positive correlation between career risks and income suggests that the reluctance to venture in one's career can potentially limit his/her earning power in the long run (King, 1974). Avoiding risky career path may reduce in one's human-capital investment that requires more entrepreneurship and initiatives, and thus dampen his/her income growth (Shaw, 1996).

However, to what extent the impaired household wealth affects personal income is ultimately an empirical question. This section analyzes the long-run impact of the career distortion arising from the household wealth shock. In particular, I estimate the sensitivity of income loss with respect to wealth damage in a flood.

[19] Based on untabulated results, the treated group is about 0.6% more likely to move than the control group after a flood relative to the pre-shock period.

The quantification exercise revolves around estimating the following model:

$$IncomeLoss_{s,i} = \alpha_s + \beta_1 PropertyDamage_{s,i} + \varepsilon_{s,i}, \tag{7}$$

where $IncomeLoss_{s,i}$ is the net present value (NPV) of the future income loss for individual i at the time of flood s, and $PropertyDamage_{s,i}$ is the nominal property damage suffered by individual i. Because individual-level property damage is not directly observable in the data, I impute property damage following the conventional method in hydrological literature (Dutta et al., 2003). The imputation process is detailed in Appendix A.4.

To maximize the power of the test, I focus on the income loss in the six and a half years after the flood, the longest horizon possible based on the LEHD data coverage. The key coefficient β_1 measures the sensitivity of an individual's income loss with respect to his/her property damage. Note the net wealth loss is likely to be smaller than the property damage with the presence of flood insurance. In this sense, β_1 should be interpreted as the lower bound of the income-wealth sensitivity.

Because the property damage one suffers is potentially endogenous to his/her income, regressing $IncomeLoss$ against $PropertyDamage$ is likely to yield a biased estimate of β_1. To address this issue, I resort to the RD design that compares residents in the 0.1-mile-wide bands on both sides of the inundation boundary, as elaborated in section 3. The RD design is likely to be well-suited for the income analysis for two reasons. First, because the treated and control groups are ex-ante indistinguishable, the control group in the RD setting is a reasonable proxy for the counterfactual states. Second, as elaborated in Appendix A.4, the inundation information available to the RD sample enables the imputation of property damage, a key data input to estimating the income-wealth sensitivity.

Having verified the identification assumption in subsection 3.1, I estimate a first-stage RD regression of the following form:

$$PropertyDamage_{s,i} = \beta_3 D_{s,i}^{Inundated} + \sum_{k=1}^{K} \gamma_k D_{s,i}^{k} + \varepsilon_{s,i}, \tag{8}$$

where all variables are defined in equations (1) and (7). Likewise, the treated (control) group

consists of individuals living within a 0.1-mile-wide band inside (outside) the flood line as of the shock. The key coefficient β_3 measures the gap between the two groups in their property damage.

The next step is to estimate the reduced-form RD specification:

$$IncomeLoss_{s,i} = \beta_4 D_{s,i}^{Inundated} + \sum_{k=1}^{K} \gamma_k D_{s,i}^{k} + \varepsilon_{s,i}, \tag{9}$$

where β_4 captures the NPV of the excess income loss for an individual in the treated group ($D_{Inundated} = 1$) relative to the control group ($D_{Inundated} = 0$). Estimating the NPV of income loss requires examining the income change around the shock. Accordingly, I run a DID regression similar to equation (1) to estimate the NPV of the income loss:

$$Income_{s,i,t} = \alpha_{s,i} + \alpha_{s,t} + \beta_5 D_{s,t}^{Post} \times D_{s,i}^{Inundated} + D_{s,t}^{Post} \times (\sum_{k=1}^{K} \gamma_k D_{s,i}^{k}) + \varepsilon_{s,i,t}, \tag{10}$$

where $Income_{s,i,t}$ is individual i's income in quarter t.[20] The analysis builds on the sample of individuals used the above RD analysis and covers the two years before and the six and a half years after the floods. β_5 represents the income loss of a treated individual in an average quarter after the flood, relative to the control group around the shock.

Panel A of Table VII reports the first-stage RD results, corresponding to equation (8). As expected, the coefficient on $D^{Inundated}$ remains significantly positive regardless of the specifications. The estimate in column (1) suggests an average treated individual suffers $12,860 in property damage, or about 14% of his/her home equity.

Panel B reports the estimation of reduced-form RD regression (i.e., equation (10)). The significant coefficient on $D^{Post} \times D^{Inundated}$ suggests household wealth loss is likely to have a lasting impact on labor income. The magnitude of the coefficient in column (1) translates into an income loss of $330 per quarter during the six and a half years after the shock, or a 5.1% decrease in earnings among the treated group.

The estimate of β_5 allows me to calculate the NPV of the income loss as a result of the

[20] This income measure covers job salary but does not include other forms of compensation such as option grants and benefits. Salary accounts for more than 90% of employment income in the U.S. according to the Bureau of Labor Statistics, which makes this measure a reasonable proxy for an individual's labor income.

wealth damage:

$$\beta_4 = \beta_5 \sum_{t=1}^{4T} \frac{1}{(1+r)^{t/4}}, \tag{11}$$

where T is the number of years in the post-period (six and a half years), and r is the discount rate. For the baseline estimate, I set the discount rate to the average five-year treasury rate during the sample period (2.325%), assuming the risk of the income loss is mostly idiosyncratic. A resulting estimate of $\beta_4 = 7,954$ indicates the treated group experiences an income loss with an NPV of \$7,954 in the six and a half years after the flood.

Combining the reduced-form and the first-stage RD results leads to an estimate of $\beta_1 = \frac{\beta_4}{\beta_3} = 0.62$ in Equation (7). In other words, \$1 of property damage can translate to an income loss of more than 60 cents in the six and a half years after the shock based on the above estimation.[21] As discussed above, this estimate should be interpreted as a lower bound of the income sensitivity to household wealth, considering the existence of partial insurance against flood damage. Overall, the above finding suggests the career distortion caused by household wealth loss has a significant effect on individuals' income in the long run.

6. Conclusion

This paper investigates how household wealth affects individuals' broadly-defined entrepreneurial career choices using a set of staggered floods in the U.S. as an exogenous wealth shock. The analysis builds on an individual-level matched dataset combining LEHD data from the U.S. Census Bureau, Corelogic deed records, and a battery of GIS flood data. Within the 0.1-mile-wide bands on both sides of the inundation boundary after a flood, the residents just below the flood line become less likely to work in startups relative to their neighbors just above the flood line. A comparison between low- and high-elevation areas within the same county extends external validity to the RD finding and confirms the damage caused by a flood has a dampening effect on individuals' desire to work for startups.

[21] Using alternative discount rates will not drastically change the magnitude of the estimate. For example, if we assume the discount rate to be the return of the S&P 500 index during the sample period, \$1 of property damage would translate to more than 50 cents of income losses in the six and a half years after the shock.

The concentration of the effect on homeowners further attributes the finding to impaired household wealth, more than a psychological impact. Additional analyses suggest the effect is likely explained by the impaired self-insurance capacity against career risks as a result of the wealth loss. The career distortion leads to a long-run personal income loss of more than 60 cents for each dollar of property damage.

Overall, the paper provides novel empirical evidence that household wealth has a pronounced, causal effect on individuals' decision to work for startups. To the best of my knowledge, the paper is the first to document this effect at the micro level. The findings help further our knowledge on the human capital of entrepreneurial firms, which is a crucial input to these firms' success. Also, the mechanism analysis and income-wealth sensitivity estimation shed light on the importance of self-insurance for entrepreneurial career choices and individuals' long-term earning power.

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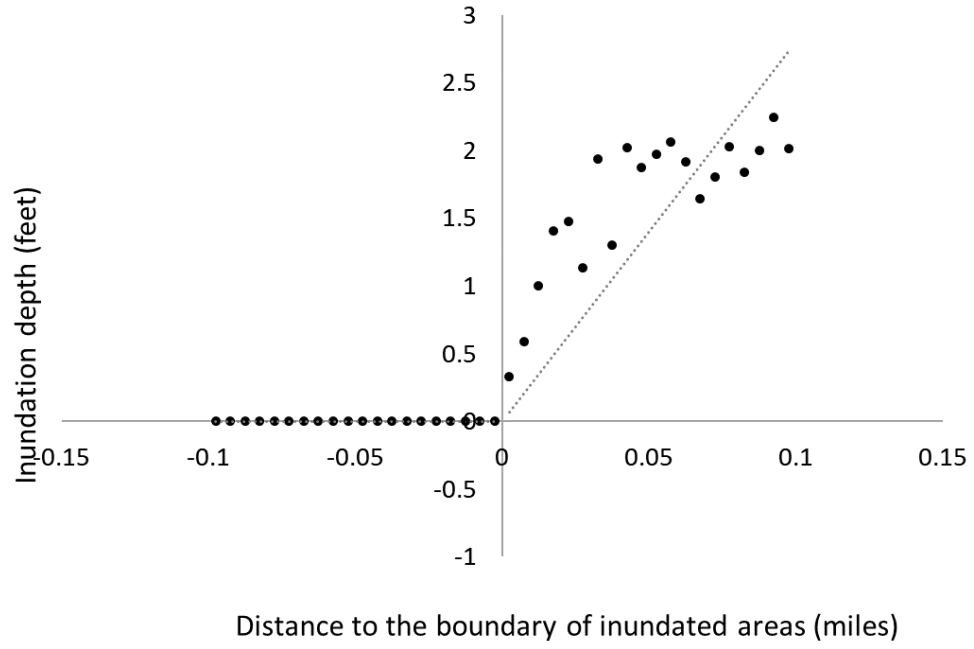
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Figure 1. Regression discontinuity analysis: Graphical evidence. Subfigure (a) shows the inundation depth of a flood around the boundary of the submerged area. The horizontal axis represents the distance between a residence location and the boundary, which is set to be negative outside the inundated area and positive inside. Each point represents the average inundation depth of all residential properties in a 0.005-mile-wide band. Subfigure (b) reports the change in individuals' entrepreneurial career choices around the boundary in a similar format to subfigure (a). Each point represents the pre-post change in the likelihood of working for startups conditional on being employed for the residents within a 0.005-mile-wide band.

(a) Inundation depth



(b) Pre-post change in entrepreneurial career choices

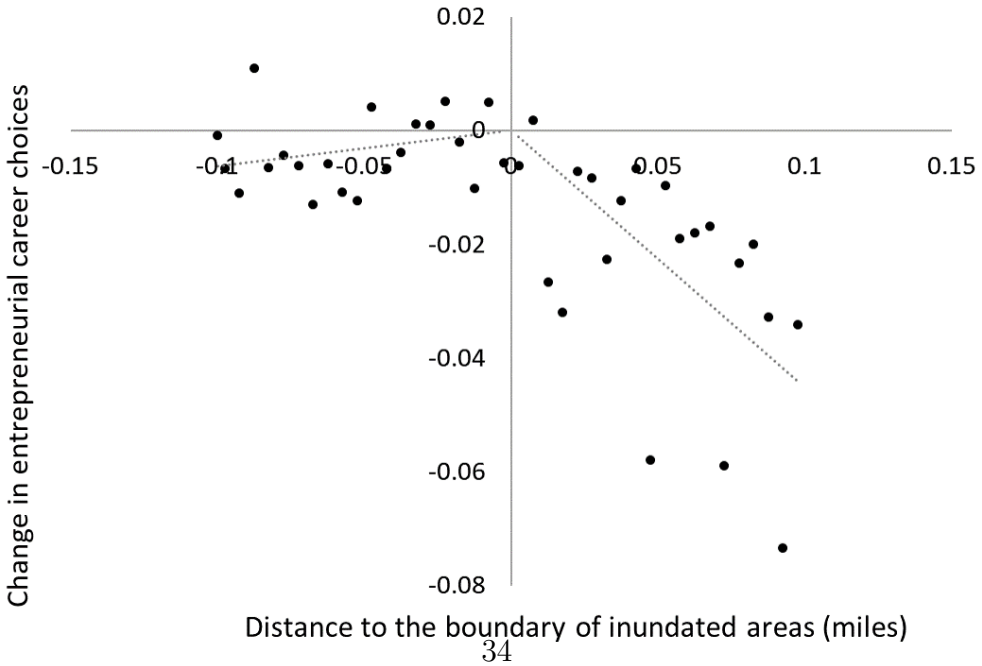


Figure 2. Parallel-trends test: Low- versus high-elevation residents. The figure reports the difference between low- and high-elevation residents on their likelihood of working in a startup around the shocks, conditional on being employed. The horizontal axis represents the quarter relative to the flood. Each point corresponds to an estimate of β_τ in equation (5). The dotted lines represent the upper and lower bounds of the 95% confidence interval of the estimates.

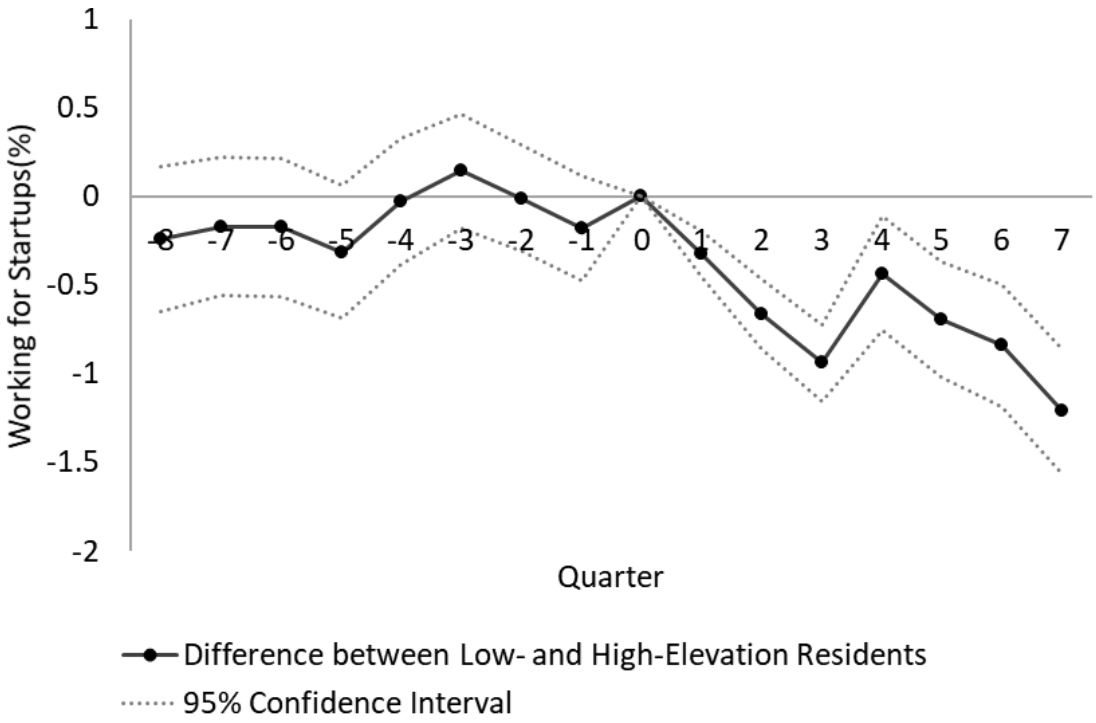


Table I Summary statistics. Panels A and B report the summary statistics of the sample used in the DID and RD analysis, respectively. *Elevation* is the elevation of an individual's residential property in feet. $D^{Startup}$ is a dummy variable equal to 1 if an individual worked in a startup in a given quarter, and 0 if he/she worked for a more mature firm. *Education* is the highest degree a person obtains on a 4-point scale (1 = *represents less than high school* and 4 = *a bachelor's or advanced degree*). *Age* is the age of an individual at the time of the flood. *Male* is a dummy variable equal to 1 for male, and 0 for female. *Income* is an individual's average quarterly income from the primary jobs over the sample period. Columns (1)–(3) report the mean, standard deviation, and median in both panels.

Panel A: DID sample

	Mean [1]	Standard Deviation [2]	Median [3]
Elevation	168.2	151.6	153.2
$D^{Startup}$	0.113	0.317	0
Education	2.713	0.992	3
Age	37.80	15.48	37
Male	0.535	0.499	1
Income	7,836	17,670	4,587

Panel B: RD sample

	Mean [1]	Standard Deviation [2]	Median [3]
$D^{Startup}$	0.101	0.306	0
Education	2.592	0.955	3
Age	38.94	14.30	38
Male	0.587	0.4924	1
Income	6,608	8,870	4,898

Table II RD design: Covariate balance. The table compares the treated and control groups in the RD analysis on predetermined demographic attributes at the individual level. *Income* is defined as the average quarterly income of an individual in the two years before the shock. $D_{Pre}^{Startup}$ is a dummy variable equal to 1 if an individual worked in a startup in a given quarter during the pre-shock period, and 0 if he/she worked for a more mature firm. *Home Value* is the nominal home value in US dollar at the time of the flood. *Home Flood Risk* (*Employer Flood Risk*) captures the ex-ante flood risk exposure of an individual's residence location (workplace). In particular, the flood risk is defined as a categorical variable on a scale of 0–2, where a value of 2 (1) indicates the residence address is within a 100-year (500-year) pre-designated flood zone, and 0 represents being out of a flood zone. *Employer Inundation Status* is a dummy variable set to 1 if in the quarter before the shock, a person worked in an establishment that would be inundated in the flood, and 0 if he/she worked outside the to-be-flooded area. *Age*, *Male*, and *Education* are defined in Table I. Columns (1)–(2) report the mean of the treated and control groups, respectively. Columns (3)–(4) present the estimated differences between the two groups and the corresponding t-statistics. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Predetermined Attributes	Treated [1]	Control [2]	Difference [3]	T-statistics [4]
Income	6,540	6,665	-124.8	(-0.57)
$D_{Pre}^{Startup}$	10.1	10.1	0.007	(0.94)
Age	38.91	38.95	-0.036	(-0.10)
Male	0.583	0.591	-0.008	(-0.64)
Education	2.583	2.599	-0.016	(-0.68)
Home Value	243,800	238,200	5,600	(1.23)
Home Flood Risk	1.173	1.159	0.014	(1.53)
Employer Flood risk	1.121	1.121	0.0002	(0.03)
Employer Inundation Status	0.160	0.142	0.018	(0.48)

Table III RD design: Regression analysis. Panels A and B are based on the individuals living in the 0.1-mile-wide bands on both sides of the boundary of the inundated area. Panel A reports the estimates of equation (1) at the individual-quarter level, where the outcome variable, $D^{Startup}$, is a dummy variable set to 1 if an individual works for a startup as a primary job during a given quarter, and 0 if he/she works for a more mature firm. D^{Post} is a dummy variable equal to 1 if a given quarter is within two years after the shock (including the quarter of the flood), and 0 if it is within two years before the shock. $D^{Inundated}$ is a dummy variable equal to 1 if an individual lived inside the inundation boundary before the shock, and 0 otherwise. Panel B corresponds to variants of equation (1), where the outcome variable, $D^{Unemployment}$, is a dummy variable equal to 1 if an individual does not have a paid job in a given quarter based on the LEHD job history, and 0 otherwise. T-statistics based on the clustered standard errors by individual are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: The decision to work for startups

Dependent variable:	$D^{Startup}$ (in %)			
	[1]	[2]	[3]	[4]
$D^{Post} \times D^{Inundated}$	-0.780***	-0.750**	-0.775**	-0.743**
	(-2.59)	(-2.55)	(-2.57)	(-2.53)
N	80,500	80,500	80,500	80,500
R-sq	0.510	0.509	0.510	0.510
Fixed effects:				
Year FE	Yes	Yes	No	No
Individual FE	Yes	Yes	Yes	Yes
Quarter FE	No	No	Yes	Yes
County × Post FE	Yes	No	Yes	No

Panel B: The likelihood of unemployment

Dependent variable:	$D^{Unemployment}$ (in %)			
	Pre-shock	Post-shock	All periods	
	[1]	[2]	[3]	[4]
$D^{Inundated}$	-0.156	-0.226		
	(-0.73)	(-1.01)		
$D^{Post} \times D^{Inundated}$			-0.08	-0.08
			(-0.27)	(-0.27)
N	41,000	43,000	84,000	84,000
R-sq	0.004	0.005	0.163	0.163
Fixed effects:				
County FE	Yes	Yes		
Year FE	Yes	Yes	Yes	Yes
Individual FE			Yes	Yes
County × Post FE			No	Yes

Table IV DID design across elevations. Panel A presents the first-stage DID analysis, and Panel B presents the reduced-form DID analysis across elevations. In Panel A, columns (1)–(4) report the estimates of equation (2) at the property level, and columns (5)–(6) report the estimates of equation (3) at the property-year level. The outcome variable in columns (1)–(2), columns (3)–(4), and columns (5)–(6) are $InunDepth$, $Damage$, and $HomeValue$, as defined in section 3.2. *Elevation* is the elevation of a property in feet. Panel B estimates equation (4) at the individual-quarter level. $D^{LowElev}$ is a dummy variable equal to 1 if before the flood, an individual lived in an elevation falling into the bottom three deciles within a county, and 0 if he/she lived at an elevation within the top three deciles. $D^{Startup}$ and D^{Post} are defined in Table III. T-statistics are reported in parentheses. In Panel B and columns (5) and (6) of Panel A (where county-level clustering is feasible), the t-statistics are based on the clustered standard errors by county. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A : Firs-stage analysis

Dependent variable:	Inundation Depth		Damage Level		Housing Value	
Housing value measure:					Nominal (in thousands)	Standardized
	[1]	[2]	[3]	[4]	[5]	[6]
Elevation	-0.033***	-0.505***	-0.008***	-0.131***		
	(-22.22)	(-64.99)	(-17.53)	(-53.76)		
D^{Post}× Elevation					3.478***	0.053***
					(4.01)	(6.32)
N	35,609	35,609	35,609	35,609	1,487,066	1,487,066
R-sq	0.110	0.471	0.102	0.452	0.993	0.943
Fixed effects:						
County FE	Yes	No	Yes	No	No	No
Zipcode FE	No	Yes	No	Yes	No	No
Post × County FE					Yes	Yes
Year FE					Yes	Yes
Property FE					Yes	Yes

Panel B: Reduced-form DID analysis

Dependent variable:	$D^{Startup}$ (in %)				
	[1]	[2]	[3]	[4]	[5]
$D^{Post} \times D^{LowElev}$	-0.487***	-0.498***	-0.524***	-0.486***	-0.500***
	(-3.39)	(-3.18)	(-3.36)	(-3.55)	(-3.19)
N	2,144,000	2,179,000	2,179,000	2,144,000	2,179,000
R-sq	0.77	0.564	0.564	0.849	0.565
Fixed effects:					
Quarter FE	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes
Post × County FE	Yes	Yes	No	No	No
Post × Firm FE	Yes	No	No	No	No
Year × County FE	No	No	No	Yes	Yes
Year × Firm FE	No	No	No	Yes	No

Table V Homeowners versus non-homeowners. This table reports the estimates of equation (6). $D_{s,i}^{Homeowner}$ is a dummy variable that indicates a homeowner identified from the matching between Corelogic and the LEHD. In particular, $D^{Homeowner}$ is set to 1 if an individual lived in his/her own property at the time of the flood and became a homeowner at least one year before the flood, and 0 otherwise. Due to the partial matching between the two datasets, the group with $D_{s,i}^{Homeowner} = 0$ is a mix of renters and unidentified homeowners. The definitions of other variables are consistent with Table IV. T-statistics based on the clustered standard errors by county are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Dependent variable:	$D^{Startup}$ (in %)	
	[1]	[2]
$D^{Post} \times D^{LowElev} \times D^{Homeowner}$	-0.351**	-0.347**
	(-2.06)	(-2.04)
$D^{Post} \times D^{LowElev}$	-0.381***	-0.368***
	(-3.03)	(-2.94)
$D^{Post} \times D^{Homeowner}$	-0.032	0.012
	(-0.26)	(0.10)
N	3,051,000	3,051,000
R-sq	0.546	0.547
Fixed effects:		
Quarter FE	Yes	Yes
Individual FE	Yes	Yes
Post × County FE	Yes	No

Table VI Self-insurance versus luxury-goods-consumption channels. The table reports the estimates of equation (4) with alternative dependent variables at the individual-quarter level. In Panel A, $D^{Startup}_{LowerIncome}$ ($D^{Startup}_{HighIncome}$) is a dummy variable equal to 1 if an individual's startup job provides lower (higher) income than his/her previous employment, and 0 if he/she works in a non-startup firm or his/her startup job comes with an income upgrade (downgrade). In Panel B, $D^{Startup}_{LowerRiks}$ ($D^{Startup}_{HigherRiks}$) is a dummy variable equal to 1 if a person's current startup job has a lower (higher) income risk than his/her previous job, and 0 otherwise conditional on being employed. In Panel C, $IndustryRisk$ measures the predetermined employment risks of the industry in which an individual works in a given quarter. T-statistics based on the clustered standard errors by county are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Upgrade versus downgrade startup jobs

Dependent variable:	$D^{Startup}_{LowerIncome}$ (in %)		$D^{Startup}_{HigherIncome}$ (in %)	
	[1]	[2]	[3]	[4]
$D^{Post} \times D^{LowElev}$	-0.171	-0.159	-0.352***	-0.339***
	(-1.54)	(-1.43)	(-2.98)	(-2.86)
N	2,179,000	2,179,000	2,179,000	2,179,000
R-sq	0.547	0.548	0.533	0.533
Fixed effects:				
Year FE	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes
County × Post FE	No	Yes	No	Yes

Panel B: High- versus low-risk startup jobs

Dependent variable:	$D^{Startup}_{LowerRisk}$ (in %)		$D^{Startup}_{HigherRisk}$ (in %)	
	[1]	[2]	[3]	[4]
$D^{Post} \times D^{LowElev}$	-0.138	-0.125	-0.374***	-0.386***
	(-1.26)	(-1.14)	(-3.10)	(-3.22)
N	2,179,000	2,179,000	2,179,000	2,179,000
R-sq	0.537	0.537	0.539	0.539
Fixed effects:				
Year FE	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes
County × Post FE	No	Yes	No	Yes

Panel C: The decision to work in high-risk industries

Dependent variable:	$IndustryRisk$ (in %)	
	[1]	[2]
$D^{Post} \times D^{LowElev}$	-0.084**	-0.078**
	(-2.11)	(-1.96)
N	2,179,000	2,179,000
R-sq	0.742	0.742
Fixed effects:		
Quarter FE	Yes	Yes
Individual FE	Yes	Yes
County × Post FE	Yes	No

Table VII Long-term implications: The sensitivity of income loss with respect to wealth damage. Panel A reports the estimates of equation (8) at the individual level. The outcome variable *Property Damage* is the nominal property damage an individual suffers due to a flood. Panel B reports the estimates of equation (10) at the individual-quarter level. The outcome variable is the labor income of an individual in a give quarter. The estimates are based on the two years before and the six and a half years after the shock. Other explanatory variables are defined in Table III. T-statistics based on the clustered standard errors by individual are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: First-stage RD analysis

Dependent variable:	Property Damage	
	[1]	[2]
$D^{Inundated}$	12,860***	13,350***
	(22.21)	(23.45)
N	6,700	6,700
R-sq	0.087	0.077
Fixed effects:		
County FE	Yes	No

Panel B: Reduced-form RD analysis

Dependent variable:	Income)	
	[1]	[2]
$D^{Post} \times D^{Inundated}$	-328.0**	-330.3**
	(-1.99)	(-2.00)
N	162,000	162,000
R-sq	0.534	0.533
Fixed effects:		
Year FE	No	Yes
Quarter FE	Yes	No
Individual FE	Yes	Yes
County × Post FE	Yes	Yes

Online Appendix for the paper

"Household Wealth and Entrepreneurial Career Choices: Evidence from Climate Disasters"

Appendix A. Regression Kink Analysis

According to Figure 1, a kink in inundation depth presents at the boundary of the flooded area. This observation suggests a regression kink design is likely to apply to this setting. To verify whether the RD finding is robust to alternative empirical methods, I estimate the kink at the threshold and report the results in Table A.1. The key independent variable is $Distance_{s,i} * D_{s,t}^{Post} * D_{s,i}^{Inundated}$, where $Distance_{s,i}$ is defined as the distance between individual i's residence location and the boundary of the inundated area, which is set to be positive inside the boundary and negative outside. The significantly negative coefficients on the triple-interaction term indicate the slope of the change in entrepreneurial career choices against the distance to the boundary is significantly larger (more negative) inside the flood line than in the narrow band outside. Considering the inundation depth is larger for the inner inundated area, this finding is consistent with the hypothesis that flood damage dampens an individual's desire to work in a startup.

Appendix B. Further Discussion about DID Analysis

A potential concern about the DID design is that the high- and low-elevation residents fundamentally differ in some characteristics. If such a characteristic happens to drive a divergence in employment decisions around the time of the flood, a change in individuals' entrepreneurial career choices may not be attributable to their wealth loss in a flood.

To address this concern, I compare the low- and high-elevation groups along a set of predetermined demographic variables, including age, gender, education, and quarterly income. The result is reported in Table A.2. Column (2) (column (3)) reports the mean of the corresponding variable for the low- (high-) elevation residents, and columns (3) and (4) present the differences between the first two columns and the t-statistics of the estimated gaps, respectively. All variables are defined in Table II. According to column (4), the two groups exhibit no significant difference except in their quarterly income in the pre-shock period.

To test whether the income discrepancy between the two groups drives the baseline

result, I incorporate $D_{s,t}^{Post} \times Income_{s,i}$ as a control variable in equation (4), where $Income_{s,i}$ is defined as the average quarterly income of individual i in the two years before the shock. If the income gap does induce the two groups to diverge in their career choices in the aftermath of the flood, I would expect the coefficient on $D_{s,t}^{Post} \times Income_{s,i}$ to be significantly positive.

Contrary to the prediction, the coefficients on $D^{Post} \times Income$ are insignificant in Panel A of Table A.3, and the coefficients on $D^{Post} * D^{LowElev}$ remain almost identical to those in Table IV. This observation suggests that the income gap is not likely to confound the DID result.

The above analysis is informative about the impact of observed characteristics, but does not rule out an unobserved confounder to the flood events. To test whether the divergence in entrepreneurial career choices is driven by an unobserved confounder coincident with the floods, I conduct a placebo test based on the counties that are not affected by the flood in the five states of interest. In particular, I examine how the entrepreneurial career choices change for the high- and low-elevation residents in these unaffected counties two years before versus two years after the five flood events. If a confounder rather than the flood leads to a divergence between the low- and high-elevation residents, I expect to observe a similar pattern in these counties even if they are not flooded. Accordingly, I reestimate equation (4) based on the unaffected counties and report the results in columns (1)–(3) of Table A.4. The insignificant coefficients on $D^{Post} \times D^{LowElev}$ indicate low- and high-elevation residents in the unaffected counties do not experience similar divergence in their decisions to work for startups, which alleviates the concern about an unobserved confounder.

To further verify the absence of a confounder, I compare the divergence across elevations in affected and unaffected counties within the same state by estimating the following regression:

$$\begin{aligned} D_{s,i,t}^{Startup} =& \alpha_{s,i} + \alpha_{s,t} + \beta_1 D_{s,t}^{Post} \times D_{s,i}^{LowElev} \times D_{s,i}^{Flood} + \beta_2 D_{s,t}^{Post} \times D_{s,i}^{LowElev} \\ &+ D_{s,t}^{Post} \times (\sum_{k=1}^{K} \gamma_k D_{s,i}^{k}) + D_{s,t}^{Post} \times (\sum_{j=1}^{J} \tau_j D_{s,i,t-1}^{j}) + \varepsilon_{s,i,t}, \end{aligned} \tag{B1}$$

where $D_{s,i}^{Flood}$ is a dummy variable equal to 1 if individual i lived in a flooded county, and 0 otherwise. $D_{s,t}^{Post}$ is a dummy variable equal to 1 for the post-period of flood event s, and 0 for

the pre-period regardless of whether an individual resides in affected or unaffected counties. Other variables are defined as in equation (4). Columns (4)–(6) of Table A.4 present the result. As expected, the coefficients on the triple-interaction term are significantly negative for all specifications, indicating the divergence between the low- and high-elevation residents in flooded counties is significantly larger that in the unaffected counties. This finding confirms that the DID effect presented in Section 3.2 is not attributed to an unobserved confounder.

Appendix C. Transition into Startups versus Retention within Startups

Two forces can potentially explain the dampening effect of wealth loss on individuals' decisions to work for startups. On the one hand, an employee of a mature firm may become less willing to join a newly established startup. On the other hand, a startup worker may decide to transition into a more mature firm if he/she suffers from wealth damage.

To explore the two potential forces, I conduct a partitioning analysis and report the results in Table A.5. The outcome variable in column (1) is $D^{Startup}_{Transition}$, a dummy variable set to 1 if an individual just transfers from an established firm into a startup and it is the first quarter he/she works in the startup, and 0 if he/she works for a more mature firm or remains in startups in a given quarter. The outcome variable in column (2) is $D^{Startup}_{Retention}$, a dummy variable equal to 1 if an individual continues to work in a startup in a given quarter, and 0 if he/she just transfers from a mature firm into a startup or is working in a non-startup firm.

The significant coefficients in columns (1) an (2) suggest both the transition rate from mature firms to startups and the retention rate within startups become lower after the wealth shock. This observation suggests the effect is not purely explained by lower churning rates as a result of the wealth shock and differentiates the finding from the dampening effects of high leverage on labor mobility (Brown and Matsa, 2019).

Also, a back-of-the-envelop calculation based on the coefficient estimate in column (1) suggests about 70.5% of the reduced likelihood of working in startups is attributed to the

lowered transitioning rate from established firms into startups, and about 29.5% of the effect is explained by an decrease in the retention rate within startups.[22]

Appendix D. Property Damage Imputation

To estimate the sensitivity of long-run income loss with respect to wealth damage, I impute property-level flood damage using a stylized model following the literature convention (e.g., Dutta et al., 2003). The model assumes flood damage to a property is proportional to the product of housing value and inundation depth. In other words,

$$PropertyDamage_i = \gamma InunDepth_i \times HousingValue_i, \tag{D1}$$

where γ is an unknown parameter to be estimated, $InunDepth_i$ denotes the depth that a property was submerged in a flood, and $HousingValue_i$ indicates the home value of individual i at the time of the flood. Following the literature convention (e.g., Gopalan et al. (2019)), the value of a property is estimated based on the historical sales price from Corelogic, adjusted by the zip-code-level housing price growth provided by the Federal Housing Finance Agency.

To estimate γ, I use the county-level property-damage information. More specifically, within the three counties where relevant data are accessible, I match the average *predicted* property damage from the above model with the average *actual* property damage retrieved from the county-level statistics:

$$\overline{PropertyDamage} = \gamma \frac{\sum_{i=1}^{I} HousingValue_i \times InunDepth_i}{I}, \tag{D2}$$

where $\overline{PropertyDamage}$ is the nominal damage of an average flooded property in the three

[22]The calculation process is as follows. Column (2) of Table IV suggests that among the treated group, the likelihood of working in a startup conditional on being employed is 0.498% lower than the pre-shock level in the two years after the flood. Suppose the rate of working in a startup declines at a constant rate during the two years. The likelihood of working in a startup in a given quarter will decrease by about $0.498\%/4.5 = 0.111\%$ from the previous quarter. As shown in column (1) of Table A.5, each quarter, the likelihood of leaving a mature firm for a startup declines by 0.0782% (conditional on being employed) compared with the pre-shock level, which accounts for $0.0782\%/0.111\% = 70.5\%$ of the baseline effect.

counties, and I denotes the number of flooded properties in the corresponding area in my sample. Because all variables except γ are available, solving equation (D2) yields an estimate of γ equal to 5.03%. In other words, one foot of inundation at the peak of a flood can lead to damage equivalent to 5.03% of the property value, comparable to the estimate in Dutta et al. (2003).

Taking γ into equation (D1) provides an estimate of individual-level property damage ($PropertyDamage_i$), which is a key data input to the estimation of the income-wealth sensitivity in equation (7).

Table A.1 Regression kink analysis. This table reports the estimates of the kink at the boundary of the inundated area. The estimates are based on the individuals living in the 0.1-mile-wide bands on both sides of the inundation boundary. *Distance* is defined as the distance between an individual's predetermined residence location to the boundary of the inundated area, set to be positive if it is inside the boundary, and negative if outside. The definitions of other explanatory variables are consistent with Table III. T-statistics based on the clustered standard errors by individual are reported in parentheses. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

Dependent variable:	$D^{Startup}$ (in %)	
	[1]	[2]
$Distance \times D^{Post} \times D^{Inundated}$	-15.06***	-15.18***
	(-3.00)	(-3.03)
$Distance \times D^{Post}$	-1.301	-1.223
	(-0.52)	(-0.49)
N	80,500	80,500
R-sq	0.51	0.51
Fixed effects:		
Individual FE	Yes	Yes
Year FE	Yes	No
Quarter FE	No	Yes
County × Post FE	Yes	Yes

Table A.2 Summary statistics by elevation. This table compares the low-elevation and high-elevation groups regarding a set of predetermined individual characteristics. Column (1) (Column (2)) reports the mean value for the low- (high-) elevation residents. Columns (3) and (4) report the differences between the two groups as well as the t-statistics of the estimated differences. All variables are defined in Table II. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Predetermined Attributes	Low-elevation [1]	High-elevation [2]	Difference [3]	T-statistics [4]
Male	0.543	0.527	0.016	(1.33)
Age	37.70	37.90	-0.197	(-0.54)
Education	2.727	2.699	0.028	(0.32)
Income	6,981	8,710	-1,729***	(-4.15)

Table A.3 DID analysis across elevations: Control for income discrepancy. The table reports the estimates of equation 4, with $D^{Post} \times Income$ as an additional control variable. *Income* denotes the average quarterly income of an individual over the two years before the shock. T-statistics based on the clustered standard errors by county are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Dependent variable:	$D^{Startup}$ (in %)		
	[1]	[2]	[3]
$D^{Post} \times D^{LowElev}$	-0.493***	-0.494***	-0.522***
	(-3.43)	(-3.15)	(-3.34)
$D^{Post}\times$ Income	-0.000	0.000	0.000
	(-0.84)	(0.53)	(0.28)
N	2,144,000	2,179,000	2,179,000
R-sq	0.770	0.564	0.564
Fixed effects:			
Quarter FE	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes
Post × County FE	Yes	Yes	No
Post × Firm FE	Yes	No	No

Table A.4 Placebo test based on unaffected counties. Columns (1)–(3) report a placebo test based on the unaffected counties in the five states of interest and correspond to equation (4). Columns (4)–(6) report the estimates of equation (B1) based on all counties in the five states, flooded or unflooded. D^{Flood} is a dummy variable equal to 1 if for the counties where the federal government declared a disaster associated with a flood, and 0 for other counties in the same state. The definitions of other variables are consistent with Table IV. T-statistics based on the clustered standard errors by county are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Dependent variable:	$D^{Startup}$ (in %)			$D^{Startup}$ (in %)		
Sample:	Unaffected Counties			Affected versus Unaffected Counties		
	[1]	[2]	[3]	[4]	[5]	[6]
$D^{Post} \times D^{LowElev} \times D^{Flood}$				-0.640**	-0.584*	-0.621*
				(-2.13)	(-1.73)	(-1.85)
$D^{Post} \times D^{LowElev}$	0.020	0.054	0.108	0.135	0.096	0.098
	(0.09)	(0.18)	(0.36)	(0.51)	(0.32)	(0.33)
N	532,000	551,000	551,000	2,675,000	2,730,000	2,730,000
R-sq	0.795	0.571	0.569	0.76	0.565	0.565
Fixed effects:						
Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes
Post × County FE	Yes	Yes	No	Yes	Yes	No
Post × Firm FE	Yes	No	No	Yes	No	No

Table A.5 Transition into startups versus retention within startups. This table reports the estimates of equation (4) using alternative outcome variables. $D^{Startup}_{Transition}$ is a dummy variable set to 1 if an individual works for a startup in a given quarter but not in the previous quarter, and 0 if he/she works for an established firm or continues to work in a startup. $D^{Startup}_{Retention}$ is defined as a dummy variable equal to 1 if an individual continues to work in a startup in a given quarter, and 0 if he/she just transitions from an established firm into a startup or is working in a non-startup. The definitions of other variables are consistent with Table IV. T-statistics based on the clustered standard errors by county are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Dependent variable:	$D^{Startup}_{Transition}$ (in %) [1]	$D^{Startup}_{Retention}$ (in %) [2]
$D^{Post} \times D^{LowElev}$	-0.0782**	-0.420***
	(-2.45)	(-2.79)
N	2,179,000	2,179,000
R-sq	0.119	0.565
Fixed effects:		
Quarter FE	Yes	Yes
Individual FE	Yes	Yes
Post × County FE	Yes	Yes